

ADNAN AHMED KHAN

AI Application Engineer | LLM Agents, Python/TypeScript, APIs & AWS

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PROFESSIONAL SUMMARY

AI and backend/platform engineer with 6+ years shipping Python, TypeScript, AWS, and distributed systems. Hands-on with LLM agents, LangGraph/LangChain workflows, tool calling, structured outputs, model routing, RAG/vector retrieval, Graphiti/Neo4j memory, and production-oriented AI automation. Built AI application runtimes and multimodal NLP workflows while owning platforms serving 50k-100k daily users and 500-800 RPS; reduced failed operations 80%, AWS spend 25%, and MTTR from about 45 minutes to under 10.

TECHNICAL SKILLS

AI / LLM: LLM applications, AI agents, multi-agent workflows, LangGraph, LangChain, CrewAI, OpenClaw, prompt/context engineering, tool calling, structured outputs, model routing, embeddings, RAG, ChromaDB, Graphiti/Neo4j

Application / Backend: Python 3.12, FastAPI, TypeScript, JavaScript, Node.js, Nest.js, Express, REST, async HTTP/httpx, microservices, workflow orchestration, WebSockets, Playwright

Cloud / Data / Reliability: AWS EKS/ECS/Lambda/RDS, Docker, Kubernetes, Terraform, Helm, GitHub Actions, ArgoCD, PostgreSQL, Redis, Kafka, Elasticsearch, MongoDB, DynamoDB, idempotency, retries, Prometheus, Grafana, Jaeger

SELECTED AI APPLICATION ENGINEERING

Verifiable Browser Agent Runtime | Python, LangGraph, FastAPI, Playwright, Chroma, Graphiti/Neo4j

- Engineered an asynchronous agent platform with grounded browser actions, semantic pre/post-condition verification, bounded recovery, and multi-agent critic/reducer reasoning.
- Built an OpenAI-style FastAPI responses gateway with streaming, tool calls, model/capability routing, retries, transcripts, embeddings, and vector retrieval; added durable workers with leases, encrypted state, replay/snapshots, and automated quality checks.

openclaw-workflow | TypeScript, Redis, MCP, JSON Schema

- Architected YAML/JSON-driven LLM pipelines with dependency graphs, parallel execution, per-step model overrides, retries/timeouts, cancellation, output gates, durable Redis/MCP state, schema validation, and partial resume.

PROFESSIONAL EXPERIENCE

HackOnTech | Senior Software Engineer, Backend & Platform | Remote / Islamabad | 2023-Present

- Own core backend/platform engineering for B2B/G2G systems serving 50k-100k daily users and 500-800 RPS peak traffic.
- Built a multimodal NLP workflow that generates video descriptions, compares them with user-provided text, and flags suspected semantic mismatches for review.
- Led a 0-to-1 overhaul into six high-throughput microservices and engineered resilient APIs/workflows with idempotency, retries, rollbacks, and recovery, cutting failed cross-service operations 80% and payment failures from about 5% to 2%.
- Reduced PostgreSQL/Redis query latency from 30+ seconds to under 1 second, reduced AWS spend 25%, and cut MTTR from about 45 minutes to under 10 with tracing and service-level observability.

Creative IT Park | Backend Developer | Islamabad | 2021-2023

- Delivered Python/FastAPI and Node.js platforms for government, university, and healthcare clients; owned production REST APIs, hardened Kafka pipelines, optimized Elasticsearch/data access, and secured healthcare ETL flows with AWS IAM.

GeekGenix | Backend Developer | 2020-2021

- Built Express/MongoDB APIs, Stripe checkout and webhook reconciliation, Passport.js authentication, and tested customer workflows.

EDUCATION

BS Computer Science - PMAS Arid Agriculture University, Rawalpindi